

Project Brief

- The Goals: Trigger FTP, SFTP, SSH, TELNET and NTP to observe behavior
- The Tools: CMD and WireShark
- The Emphasis: Encrypted protocols vs. Unencrypted protocols
- Methodology: CMD to give commands, Screenshots to document and WireShark to capture the protocols “red handed”

Setup

- The command:

pkgmgr /iu:"TelnetClient"

1) Pkgmgr=Windows package manager (windows tool to install/remove features)

2) /iu=Install update

3) "TelnetClient"=The feature to install-Telnet

- The command: **ftp /?** = Shows ftp details if ftp exists

- Double check **Telnet** is on

```
Administrator: שורת הפקודה
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

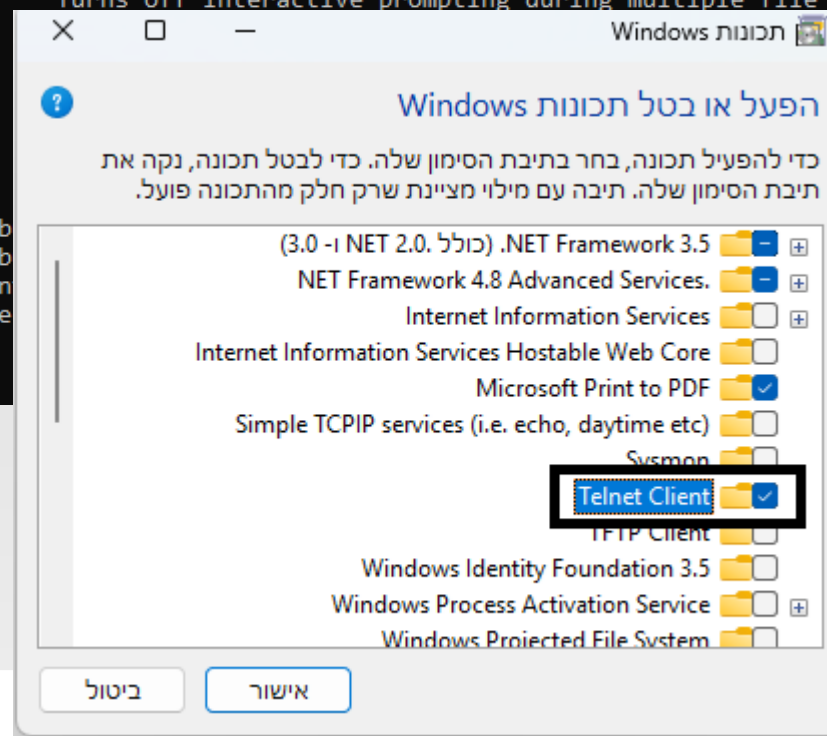
C:\Windows\System32 pkgmgr /iu:"TelnetClient"

C:\Windows\System32 ftp /?

Transfers files to and from a computer running an FTP server service
(sometimes called a daemon). Ftp can be used interactively.

FTP [-v] [-d] [-i] [-n] [-g] [-s:filename] [-a] [-A] [-x:sendbuffer] [-r:
host]

-v          Suppresses display of remote server responses.
-n          Suppresses auto-login upon initial connection.
-i          Turns off interactive prompting during multiple file
-d          Suppresses display of remote server responses.
-g          Suppresses auto-login upon initial connection.
-s:filename Turns off interactive prompting during multiple file
-a          Suppresses display of remote server responses.
-A          Suppresses auto-login upon initial connection.
-x:send sockb Turns off interactive prompting during multiple file
-r:recv sockb Suppresses display of remote server responses.
-b:async coun Suppresses auto-login upon initial connection.
-w:window size Turns off interactive prompting during multiple file
host          Suppresses display of remote server responses.
```



FTP (File Transfer Protocol) – Part 1

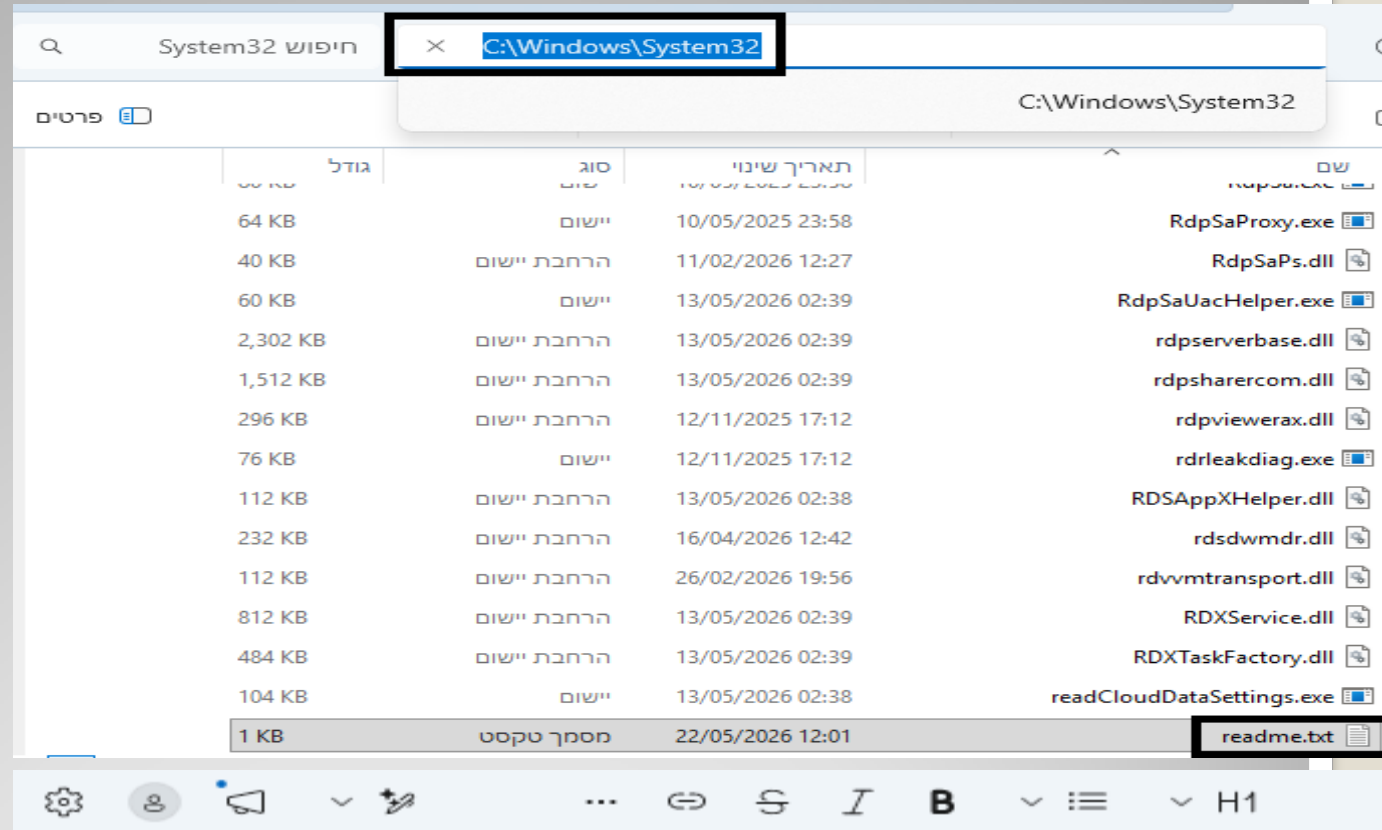
- The command: **ftp test.rebex.net** = connect through FTP to the server test.rebex.net
- Next I logged in as **anonymous**
- Then **set password to MY EMAIL** (not visible in CMD but we will see it in wireshark)
- The command **ls**=list all files in directory. The file **readme.txt** appears
- The command **get readme.txt** downloads the file

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ftp test.rebex.net
Connected to test.rebex.net.
220-Welcome to test.rebex.net!
      See https://test.rebex.net/ for more information and terms of use.
220 If you don't have an account, log in as 'anonymous' or 'ftp'.
200 Enabled UTF-8 encoding.
User (test.rebex.net:(none)): anonymous
331 Anonymous login OK, send your complete email address as your password.
Password: 
230 User 'anonymous' logged in.
ftp> ls
200 PORT OK.
150 Opening 'BINARY' data connection.
pub
readme.txt
226 Transfer complete.
ftp: 20 bytes received in 0.00Seconds 20.00Kbytes/sec.
ftp> get readme.txt
200 PORT OK.
150 Opening 'BINARY' data connection.
226 Transfer complete.
ftp: 379 bytes received in 0.00Seconds 379000.00Kbytes/sec.
ftp> bye
221 Closing session.
```

FTP – Part 2

- The file is downloaded to the folder where CMD is installed



- Opening the file readme.txt

Welcome to test.rebex.net!

You are connected to an FTP or SFTP server used for testing purposes by Rebex FTP/SSL or Rebex SFTP sample code. Only read access is allowed.

For information about Rebex FTP/SSL, Rebex SFTP and other Rebex libraries for .NET, please visit our website at <https://www.rebex.net/>

For feedback and support, contact support@rebex.net

Thanks!

FTP (File Transfer Protocol) – Part 3 - WireShark

ftp

Info	Length	Protocol	Destination	Source	Time
!Response: 220-Welcome to test.rebex.net 229		FTP	192.168.1.5	194.108.117.16	0.724220000 9
Request: OPTS UTF8 ON 68		FTP	194.108.117.16	192.168.1.5	0.741357300 10
.Response: 200 Enabled UTF-8 encoding 87		FTP	192.168.1.5	194.108.117.16	0.822081500 11
Request: USER anonymous 70		FTP	194.108.117.16	192.168.1.5	7.825260500 74
.Response: 331 Anonymous login OK, send your complete email address as your password 134		FTP	192.168.1.5	194.108.117.16	7.906044200 75
Request: PASS avivnofar@gmail.com 80		FTP	194.108.117.16	192.168.1.5	14.768389700 88
.Response: 230 User 'anonymous' logged in 91		FTP	192.168.1.5	194.108.117.16	14.850744200 89
Request: PORT 192,168,1,5,210,78 79		FTP	194.108.117.16	192.168.1.5	33.493322800 270
.Response: 200 PORT OK 72		FTP	192.168.1.5	194.108.117.16	33.575414600 271
Request: NLST 60		FTP	194.108.117.16	192.168.1.5	33.577261000 272
.Response: 150 Opening 'BINARY' data connection 97		FTP	192.168.1.5	194.108.117.16	33.657814000 275
.Response: 226 Transfer complete 82		FTP	192.168.1.5	194.108.117.16	33.748180200 282
Request: PORT 192,168,1,5,210,79 79		FTP	194.108.117.16	192.168.1.5	45.869220300 298
.Response: 200 PORT OK 72		FTP	192.168.1.5	194.108.117.16	45.949711000 299
Request: RETR readme.txt 71		FTP	194.108.117.16	192.168.1.5	45.954333100 300
.Response: 150 Opening 'BINARY' data connection 97		FTP	192.168.1.5	194.108.117.16	46.032756600 303
.Response: 226 Transfer complete 82		FTP	192.168.1.5	194.108.117.16	46.120632500 310
Request: QUIT 60		FTP	194.108.117.16	192.168.1.5	56.235276100 338
.Response: 221 Closing session 80		FTP	192.168.1.5	194.108.117.16	56.315694300 339

- Shockingly, the username AND PASSWORD appears on the screen unencrypted, a negative feature of FTP

Request: PASS avivnofar@gmail.com

- Filtering WireShark to ftp-data reveals:

- The ls command to list files (1st line, NLST)
- The actual file download (2nd line, RETR readme.txt)

ftp-data

Info	Length	Protocol	Destination	Source	Time
FTP Data: 17 bytes (PORT) (NLST) 75		FTP-DATA	192.168.1.5	194.108.117.16	33.665442300 277
FTP Data: 379 bytes (PORT) (RETR readme.txt) 437		FTP-DATA	192.168.1.5	194.108.117.16	46.038024100 305

SSH/SFTP (Secure Shell/Secure FTP)

```
C:\Windows\System32 ssh demo@test.rebex.net
The authenticity of host 'test.rebex.net (194.108.117.16)' can't be established.
ED25519 key fingerprint is SHA256:d7Te2DHmvBNSWJNBWIk2KbDTjmWtYHe2bvXTMM9lVg4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'test.rebex.net' (ED25519) to the list of known hosts.
Welcome to test.rebex.net! See https://test.rebex.net/ for more information.
(demo@test.rebex.net) Password:
For a list of supported commands, type 'help'.
demo@test:/$
```

- Next we're going to approach the same server with a secured protocol, SSH, to have a remote encrypted connection
- Note: SFTP is running on SSH, so in this test we are in fact using SFTP to download (the same file)
- The command: **ssh demo@test.rebex.net**. Demo is the username to log in as to the server. Note that the login is to the same server as we did with FTP (intentionally designed this way for testing the different protocols)
- Lines 2-4: We can see layers of security. The key is the encryption and the fingerprint is the "server id" (like MAC number). In line 4 we are specifically asked if we want to connect.
- Next type password (shows blank in CMD for security, like in the FTP process)
- The last line: demo@test:/\$ - established connection. Demo is the username and test is the server name

SSH/SFTP (Wireshark)

- As we can see the entire communication caught in WireShark is **encrypted** (filter ssh) while in FTP protocol even the password was visible. That's the main reason for SFTP.

Client: Protocol (SSH-2.0-OpenSSH_for_Windows_9.5)	87	SSHv2	194.108.117.16	192.168.1.5	1.013468100	16
Server: Protocol (SSH-2.0-RebexSSH_8.0.0.0)	84	SSHv2	192.168.1.5	194.108.117.16	1.088594100	18
Server: Key Exchange Init	1090	SSHv2	192.168.1.5	194.108.117.16	1.091087000	19
Client: Key Exchange Init	1486	SSHv2	194.108.117.16	192.168.1.5	1.091220800	20
Client: Elliptic Curve Diffie-Hellman Key Exchange Init	102	SSHv2	194.108.117.16	192.168.1.5	1.214124100	23
Server: Elliptic Curve Diffie-Hellman Key Exchange Reply	250	SSHv2	192.168.1.5	194.108.117.16	1.297986800	41
Server: New Keys	74	SSHv2	192.168.1.5	194.108.117.16	1.297986800	42
Server: Encrypted packet (len=420)	478	SSHv2	192.168.1.5	194.108.117.16	1.297986800	43
Client: New Keys	70	SSHv2	194.108.117.16	192.168.1.5	4.788460000	87
Client: Encrypted packet (len=44)	98	SSHv2	194.108.117.16	192.168.1.5	4.917990100	89
Server: Encrypted packet (len=44)	102	SSHv2	192.168.1.5	194.108.117.16	4.998856100	90
Client: Encrypted packet (len=60)	114	SSHv2	194.108.117.16	192.168.1.5	4.999229100	91
Server: Encrypted packet (len=116)	174	SSHv2	192.168.1.5	194.108.117.16	5.077529500	92
Server: Encrypted packet (len=60)	118	SSHv2	192.168.1.5	194.108.117.16	5.077529500	93
Client: Encrypted packet (len=84)	138	SSHv2	194.108.117.16	192.168.1.5	5.078473700	95
Server: Encrypted packet (len=60)	118	SSHv2	192.168.1.5	194.108.117.16	5.160538100	96
Client: Encrypted packet (len=84)	138	SSHv2	194.108.117.16	192.168.1.5	14.505953500	110
Server: Encrypted packet (len=28)	86	SSHv2	192.168.1.5	194.108.117.16	14.587823500	111
Client: Encrypted packet (len=52)	106	SSHv2	194.108.117.16	192.168.1.5	14.593324600	112
Server: Encrypted packet (len=44)	102	SSHv2	192.168.1.5	194.108.117.16	14.676752400	113
Client: Encrypted packet (len=136)	190	SSHv2	194.108.117.16	192.168.1.5	14.677420700	114
Server: Encrypted packet (len=36)	94	SSHv2	192.168.1.5	194.108.117.16	14.755835100	115
Server: Encrypted packet (len=36)	94	SSHv2	192.168.1.5	194.108.117.16	14.755835100	116
Server: Encrypted packet (len=36)	94	SSHv2	192.168.1.5	194.108.117.16	14.755835100	117
Server: Encrypted packet (len=84)	142	SSHv2	192.168.1.5	194.108.117.16	14.755835100	118
Server: Encrypted packet (len=52)	110	SSHv2	192.168.1.5	194.108.117.16	14.756166400	120
Client: Encrypted packet (len=68)	122	SSHv2	194.108.117.16	192.168.1.5	20.410919500	122
Client: Encrypted packet (len=68)	122	SSHv2	194.108.117.16	192.168.1.5	22.550508300	124
Client: Encrypted packet (len=68)	122	SSHv2	194.108.117.16	192.168.1.5	24.480403000	148

Telnet

towel.blinkenlights.nl:

Command explained:

telnet: The Telnet client —
raw unencrypted remote
connection

The hostname of a famous public Telnet server

The output: various Star Wars stories and animations

```
C:\Windows\System32>telnet towel.blinkenlights.nl
```



`C:\> Telnet towel.blinkenlights.nl`

[illegible]

```
0:~$ Telnet towel.blinkenlights.nl
```

Telnet – More Animations

```
C:\Windows\System32>telnet towel.blinkenlights.nl
```

Telnet towel.blinkenlights.nl

blinkenlights.nl



- Quitting Telnet blinkenlights

1) ctrl+]

2) quit

Administrator: שורת הפקודה

Welcome to Microsoft Telnet Client

Escape Character is 'CTRL+']'

Microsoft Telnet> quit

C:\Windows\System32>

Telnet – Wireshark – Every packet is visible

The problem is that anyone capturing traffic on the same network can see the data in the packets too

The image displays two screenshots of the Wireshark network protocol analyzer, illustrating the capture of Telnet traffic. Both screenshots show a packet list on the left, a packet details pane in the middle, and a packet bytes pane on the right.

Top Screenshot:

- Packet List:** Shows several Telnet packets. The selected packet is a Telnet packet (No. 169) with a length of 1518 bytes, containing data from 192.168.1.5 to 213.136.8.188.
- Packet Details:** Shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Telnet.
- Packet Bytes:** Displays the raw data of the Telnet packet, which includes ASCII art. The ASCII art depicts a character with a large head, wearing a hat and a suit, holding a cane.

Bottom Screenshot:

- Packet List:** Shows a different set of Telnet packets. The selected packet is a Telnet packet (No. 736) with a length of 1046 bytes, containing data from 192.168.1.5 to 213.136.8.188.
- Packet Details:** Shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Telnet.
- Packet Bytes:** Displays the raw data of the Telnet packet, which includes ASCII art. The ASCII art depicts a character with a large head, wearing a hat and a suit, holding a cane.

Telnet vs. SSH

- Next we're going to use SSH for a secure remote connection (in contrast to Telnet)
- Given a few commands to trigger connection
- The command **pwd** shows the current path
- The command **whoami** shows the current user identity (demo) to force the system to communicate with the database

C:\> Administrator: שורת הפקודה

```
C:\Windows\System32>ssh demo@test.rebex.net
Welcome to test.rebex.net! See https://test.rebex.net/ for more information.
(demo@test.rebex.net) Password:
For a list of supported commands, type 'help'.
demo@test:/$ ls
pub
readme.txt
demo@test:/$ pwd
/
demo@test:/$ whoami
demo
demo@test:/$ exit
Disconnecting...
Connection to test.rebex.net closed.

C:\Windows\System32>_
```

Telnet vs. SSH - Wireshark

- As expected everything in SSH protocol is encrypted

- Deep dive:

Lines 43-44 – the client and the server agrees on SSH version 2

Lines 45, 47 - Both sides sending a list of encryption algorithms they support — negotiating which ones to use.

The large packet sizes (1090 and 1486 bytes) show how many options are being exchanged.

Lines 49-50 – The two sides agree on the encryption and exchange keys

Info	Length	Protocol	Destination	Source	Time	
Client: Protocol (SSH-2.0-OpenSSH_for_Windows_9.5) 87		SSHv2	194.108.117.16	192.168.1.5	1.288725500	43
Server: Protocol (SSH-2.0-RebexSSH 8.0.0.0) 84		SSHv2	192.168.1.5	194.108.117.16	1.367677300	44
Server: Key Exchange Init 1090		SSHv2	192.168.1.5	194.108.117.16	1.372143500	45
Client: Key Exchange Init 1486		SSHv2	194.108.117.16	192.168.1.5	1.390751900	47
Client: Elliptic Curve Diffie-Hellman Key Exchange Init 102		SSHv2	194.108.117.16	192.168.1.5	1.530137800	49
Server: Elliptic Curve Diffie-Hellman Key Exchange Reply 250		SSHv2	192.168.1.5	194.108.117.16	1.623810700	50
Server: New Keys 74		SSHv2	192.168.1.5	194.108.117.16	1.623810700	51

Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface \Device\NPF_{A1122FF9-242E-4C28-BFD5-A651F1EA82A0}, id 0
Ethernet II, Src: LiteonTechno_f3:09:f1 (30:d1:6b:f3:09:f1), Dst: IskratelDoo_21:1e:50 (54:c2:50:21:1e:50)
Internet Protocol Version 4, Src: 192.168.1.5, Dst: 194.108.117.16
Transmission Control Protocol, Src Port: 64792, Dst Port: 22, Seq: 1466, Ack: 1059, Len: 48
SSH Protocol
SSH Version 2 (encryption:chacha20-poly1305@openssh.com mac:<implicit> compression:none) <
[Direction: Client to Server]

So Which Protocol is Secured?

Protocol	Default Port	Encrypted
Telnet	23	☐ No
SSH	22	☐ Yes
FTP	21	☐ No
SFTP	22	☐ Yes
HTTP	80	☐ No
HTTPS	443	☐ Yes

NTP (Network Time Protocol) - CMD

- Step 1: Activate

Windows Time Service

- Step 2: Start recording on

WireShark

- Step 3: force Windows to resync the clock to trigger the protocol (otherwise it would trigger

automatically every 17 minutes)

- Step 4: Check status

•Note: NTP operates on **port 123** and **UDP** network protocol

```
C:\Windows\System32>net start "Windows Time"
The Windows Time service is starting.
The Windows Time service was started successfully.
```

```
C:\Windows\System32>w32tm /resync /force
Sending resync command to local computer
The command completed successfully.
```

```
C:\Windows\System32>
```

```
C:\Windows\System32>w32tm /query /status
Leap Indicator: 0(no warning)
Stratum: 4 (secondary reference - syncd by (S)NTP)
Precision: -23 (119.209ns per tick)
Root Delay: 0.0787234s
Root Dispersion: 8.1993319s
ReferenceId: 0x14653909 (source IP: 20.101.57.9)
Last Successful Sync Time: 22/05/2026 13:35:18
Source: time.windows.com,0x9
Poll Interval: 10 (1024s)
```

```
C:\Windows\System32>_
```

NTP - WireShark

- The WireShark capture tells a straightforward story of 2 packets: 1st from the client asking for the time, and then a 2nd from the server responding with the current time
- OSI/TCP model: Ethernet 2, src/dst MAC address (layer 2). Then, internet protocol and src/dst IP addresses (layer 3). Then UDP protocol and logical port 123 (layer 4). And finally, NTP version 3 on the application layer (layers 5-7)

Info	Length	Protocol	Destination	Source	Time
NTP Version 3, client 90		NTP	20.101.57.9	192.168.1.5	3.586276100 4→
NTP Version 3, server 94		NTP	192.168.1.5	20.101.57.9	3.671318700 5←

Frame 4: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface \Device\NPF_{A1122FF9-242E-4C28-BFD5-A651F1EA82A0}, id
Ethernet II, Src: LiteonTechno f3:09:f1 (30:d1:6b:f3:09:f1), Dst: Iskrateldoo 21:1e:50 (54:c2:50:21:1e:50)
Internet Protocol Version 4, Src: 192.168.1.5, Dst: 20.101.57.
User Datagram Protocol, Src Port: 123, Dst Port: 123
Network Time Protocol (NTP Version 3, client
Flags: 0xdb, Leap Indicator: unknown (clock unsynchronized), Version number: NTP Version 3, Mode: client
[Response In: 5]
Peer Clock Stratum: unspecified or invalid (0)
Peer Polling Interval: 10 (1024 seconds)
Peer Clock Precision: -23 (0.000000119 seconds)
Root Delay: 0.078720 seconds
Root Dispersion: 9.199326 seconds
Reference ID: NULL

THANK YOU FOR WATCHING